STM32G431KBTx USB-C Sensor Module Project Report

This document summarizes the PCB Designer Intern Assessment project completed using KiCad. The project involves designing a compact STM32G431KBTx based USB-C sensor interface module with UART, I2C, and SWD debugging connectivity.

1. Project Overview

The designed PCB module is based on the STM32G431KBTx microcontroller. The board includes:
• USB Type-C connector for power and USB data communication
• SWD debugging interface
• I2C sensor connector
• UART communication connector
• Status LED indication circuit
• Power filtering capacitors and pull-up resistors

2. Purpose and Applications

The purpose of this project is to create a compact embedded development and sensor interface board for embedded firmware development, sensor interfacing, UART communication testing, USB powered embedded systems, and debugging using SWD.

3. Design Software and Libraries Used

Category	Library Type
Schematic Symbols	Standard KiCad Symbol Libraries
PCB Footprints	Standard KiCad Footprint Libraries
PCB Design Tool	KiCad PCB Editor
Gerber Generation	KiCad Plot and Drill Tools

4. Footprints Used

Component	Footprint
STM32G431KBTx	LQFP-32_7x7mm_P0.8mm
USB-C Connector	USB_C_Receptacle_GCT_USB4085
Resistors	R_0805_2012Metric
Capacitors	C_0805_2012Metric
LED	LED_0805_2012Metric
Headers	PinHeader_1x04_P2.54mm_Vertical

5. Bill of Materials (BOM)

Reference	Qty	Value	Footprint
C1, C2	2	100nF	C_0805_2012Metric

D1	1	STATUS_LED	LED_0805_2012Metric
J1	1	USB Type-C	USB_C_Receptacle_GCT_USB4085
J2	1	SWD Header	PinHeader_1x04_P2.54mm_Vertical
J3	1	I2C Header	PinHeader_1x04_P2.54mm_Vertical
J4	1	UART Header	PinHeader_1x04_P2.54mm_Vertical
R1, R2	2	5.1k	R_0805_2012Metric
R3	1	330	R_0805_2012Metric
R4, R5	2	4.7k	R_0805_2012Metric
U1	1	STM32G431KBTx	LQFP-32_7x7mm_P0.8mm

6. Conclusion

The STM32G431KBTx USB-C Sensor Module was successfully designed in KiCad including schematic capture, PCB layout, routing, copper pour, DRC validation, Gerber generation, and BOM generation. The project demonstrates embedded hardware design skills and PCB design workflow understanding.